Notes: Favilukis, Mabille & Van Nieuwerburgh (RES, 2023)
Affordable Housing and City Welfare

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1 Big picture

- **Question.** How should cities evaluate rent stabilization, upzoning, housing vouchers, and cash transfers when those policies change rents, prices, construction, labour supply, migration, and risk sharing simultaneously?
- **Core framework.** The paper builds a dynamic stochastic spatial equilibrium model with two metros, two zones inside the gateway city, overlapping generations, incomplete markets, housing tenure choice, labour supply, migration, endogenous rents and house prices, and local construction.
- **Calibration.** The gateway metro is calibrated to the New York MSA. Zone 1 is Manhattan; Zone 2 is the rest of the MSA; the outside metro is calibrated to the average of the next 74 largest US MSAs.
- **Central mechanism.** Affordable housing policies provide insurance against income and housing-risk shocks, especially for low-income households, but they distort housing supply, spatial sorting, labour supply, and migration.
- **Main result.** Expanding rent stabilization raises average welfare by 0.91% in consumption-equivalent variation because insurance gains dominate distortions.
- **Policy-design result.** Income qualification for rent stabilization improves targeting and yields a 0.66% welfare gain.
- **General-equilibrium result.** Vouchers can help direct recipients but generate no average welfare gain when financed by distortionary taxes and when migration adjusts.
- **Bottom line.** The paper reframes affordable housing as insurance in an incomplete-markets spatial economy, not just as a rent-control or housing-supply problem.

2 Incremental contribution of the paper

- **Connects macro-finance and urban economics.** It adds spatial location choice and housing policy to a heterogeneous-agent life-cycle model with endogenous prices, rents, wages, wealth, housing, mortgages, and migration.
- **Adds risk and insurance to spatial policy analysis.** Risk aversion and incomplete markets make affordable housing valuable because it stabilizes marginal utility for vulnerable households.
- **Quantifies general equilibrium policy trade-offs.** The model jointly measures supply responses, price/rent changes, labour-supply distortions, commuting, spatial sorting, and inter-city migration.
- **Compares realistic instruments.** The paper compares RS expansion, income qualification, moving RS to suburbs, upzoning, vouchers, and progressive cash transfers using a common welfare criterion.
- **Shows why financing matters.** The interaction between distortionary taxation and migration is first order: a voucher expansion funded by income taxes can fail even when the direct transfer helps poor households.
- **Gives an insurance interpretation of rent stabilization.** RS is evaluated as a partial substitute for missing insurance markets, not only as a source of landlord and tenant distortions.

3 Data and measurement

3.1 Spatial units and outside option

- Zone 1 is Manhattan, interpreted as the urban core with higher productivity, amenities, density, and housing costs.

- Zone 2 is the rest of the New York MSA.
- The outside MSA represents the rest of the country and is calibrated to the average of the next 74 largest US MSAs.

Interpretation: The two-zone structure is compact but captures the central city trade-off: high productivity and amenities in the core versus cheaper housing outside.

3.2 Households and migration

- The model targets population shares, average age, share over 65, income distribution, commuting costs, and migration rates by age and income.
- Migration is measured as in-migration into New York and out-migration from New York.
- Moving costs are calibrated to match migration by age and income.

Interpretation: Migration is not a side feature. It determines whether local housing policies cause exit of productive households or entry of households seeking insurance.

3.3 Housing prices, rents, and affordability

- Table 1 reports market prices, rents, price-to-income ratios, rent-to-income ratios, homeownership, unit size, and the prevalence of RS.
- The calibration targets key New York facts: high housing costs, high rent burden, and large spatial differences between Manhattan and the rest of the metro.
- Rent burden is central because it motivates the policy experiments.

Interpretation: A credible policy model must reproduce the affordability problem before being used for counterfactual policy evaluation.

3.4 Rent stabilization measurement

- Rent stabilization is modelled as a mandated share of rental square footage whose rent grows more slowly than market rent.
- In stationary equilibrium, slower rent growth becomes a rent discount that grows with tenure.
- RS units are allocated by lottery, have a maximum size constraint, and have no income qualification in the benchmark.
- **Access to insurance** is the probability that a lower-income household with a negative productivity shock obtains RS housing.
- **Stability of insurance** is the probability that a low-income household already in RS remains in the system next period.

Interpretation: The paper measures affordable housing not only by rent levels, but also by the probability of getting and keeping insurance-like housing stability.

4 Model and theoretical specifications

4.1 Household dynamic problem

Households choose location, tenure, housing consumption, non-housing consumption, labour supply, savings, mortgage debt, and migration. A compact recursive representation is:

$$V(s) = \max_{c,h,n,a',\ell,tenure} \{u(c,h,n,\ell) + \beta E[V(s')]\}.$$

Interpretation: The state vector includes age, productivity, wealth, housing tenure, location, RS status, and shocks. Dynamic choice matters because policy affects future rents, mobility, and the value of housing stability.

4.2 Labour income and agglomeration

$$y_t^{lab} = W_m \times A_\ell \times G_a \times z_t \times n_t.$$

Interpretation: The productivity shifter A_ℓ captures agglomeration. If affordable housing changes who lives in Manhattan, it can change total effective labour supply and output.

4.3 Rent-stabilized housing

$$R_{\ell,d}^{RS} = \kappa_1(d) R_\ell^{mkt}, \qquad \kappa_1(d) < 1,$$

where d is tenure in the RS unit. The rent discount increases with tenure.

Interpretation: This feature creates stability for incumbent tenants, but it also lowers turnover and can allocate scarce RS units to households that are no longer the neediest.

4.4 Housing supply and zoning

Construction costs rise as the housing stock approaches the zone-specific limit:

$$MC_\ell(H_\ell) \text{ increases as } H_\ell \to \bar{H}_\ell.$$

Interpretation: Zoning is a capacity constraint. Upzoning increases $\bar{H}_1$, lowers core rents and prices, and changes who wants to live in the urban core.

4.5 Welfare criterion

The main welfare measure is a consumption-equivalent variation for households living in New York before the reform:

CEV = uniform consumption change making pre-reform NY residents indifferent between benchmark and reform.

Interpretation: Fixing the pre-reform NY population avoids mechanically changing welfare by changing who moves into the city after the reform.

5 Calibration, identification, and theoretical design

- **Nature of evidence.** This is a calibrated structural policy paper, not a reduced-form causal design.
- **Discipline from data.** The calibration matches demographics, migration, income, housing size, prices, rents, ownership, affordability, and RS prevalence.
- **Internal validation.** Figures 1–4 and Table 1 show that the baseline economy reproduces key targeted and non-targeted patterns.
- **Counterfactual logic.** Policy reforms change policy rules or constraints; the economy is solved again with rents, prices, construction, labour supply, and migration adjusting endogenously.
- **General equilibrium discipline.** A policy can help recipients but lower welfare if it contracts housing supply, raises taxes, pushes high-productivity households out, or reduces effective labour supply.
- **Role of migration.** Migration is a major adjustment margin. The paper compares open-city outcomes to no-migration counterfactuals in the appendix to isolate this channel.
- **Main limitation.** Welfare effects depend on model credibility: preferences, moving costs, risk aversion, housing supply elasticities, zoning limits, RS allocation, and tax financing.

Interpretation: The results should be read as quantitatively disciplined structural counterfactuals rather than direct quasi-experimental treatment effects.

6 Tables and figures

6.1 Figure 1: Inter-MSA migration rates by age and income; Table 1: New York metro data targets and model fit

Read: Figure 1 compares out-migration from New York and in-migration into New York by age and income. Table 1 reports the model fit for population, income, ownership, prices, rents, affordability, and RS share.

Detailed interpretation: These two visuals are the credibility foundation for the model. If the model did not match migration and housing prices, counterfactuals about rent stabilization and vouchers would be unreliable. The migration fit matters because welfare effects are dampened or amplified when households can vote with their feet.

Economic message: The baseline model is plausible enough to use as a policy laboratory: it reproduces high housing costs, rent burden, spatial inequality, and migration patterns in the New York MSA.

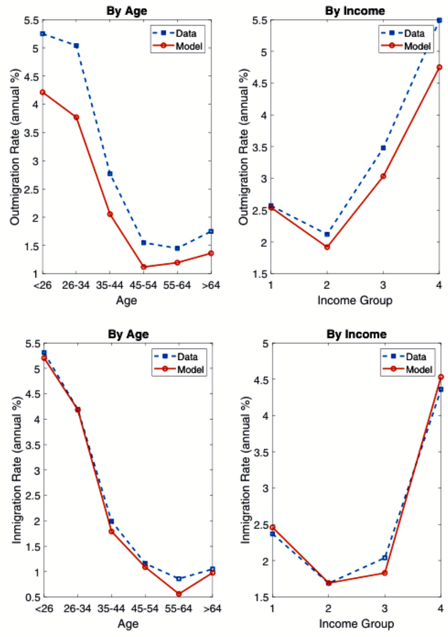


Figure 1

(a) Figure 1: Inter-MSA migration rates by age and income. The figure validates the moving-cost calibration by comparing model and data migration patterns.

TABLE 1
New York metro data targets and model fit

		Data		Model	
		Metro	Ratio Zone 1/Zone 2	Metro	Ratio Zone 1/Zone 2
1	Households (thousands)	7124.9	0.12	7124.9	0.12
2	Avg. hh age, cond. age > 20	47.6	0.95	**46.5**	**0.86**
3	People over 65 as % over 20	19.1	0.91	**19.5**	0.90
4	Avg. house size (sqft)	1644	0.59	1644	**0.63**
5	Avg. pre-tax lab income ($)	124091	1.66	124165	1.69
6	Home ownership rate (%)	51.5	0.42	**58.4**	**0.57**
7	Median mkt price per unit ($)	510051	3.11	**496649**	**2.25**
8	Median mkt price per sqft ($)	353	5.24	**280**	**3.55**
9	Median mkt rent per unit (monthly $)	2390	1.65	**2471**	**1.76**
10	Median mkt rent per sqft (monthly $)	1.65	2.78	**1.39**	2.77
11	Median mkt price/median mkt rent (annual)	17.79	1.89	16.75	**1.28**
12	Mkt price/avg. income (annual)	3.99	1.86	**4.00**	**1.33**
13	Avg. rent/avg. income (%)	24.0	1.00	23.9	**1.04**
14	Avg. rent/income ratio for renters (%)	42.1	0.81	**33.9**	**1.21**
15	Rent burdened (%)	53.9	0.79	**54.7**	**1.32**
16	% RS of all housing units	14.63	3.11	14.20	3.28

Notes: Columns 2–3 report the values for the data of the variables listed in the first column. Data sources and construction are described in detail in Supplementary Appendix B. Column 3 reports the ratio of the Zone 1 value to the Zone 2 value in the data. Columns 4 and 5 are the corresponding moments in the model. Moments in boldface are not directly targeted by the calibration.

home ownership over the life-cycle. Table 1 shows some key moments; moments in boldface are

(b) Table 1: New York metro data targets and model fit. The table reports targeted and non-targeted moments for the metro and Zone 1/Zone 2 ratios.

6.2 Figure 2: Geographic distribution of households by productivity; Figure 3: Income, wealth, and home ownership over the life cycle

Read: Figure 2 shows productivity sorting by zone and tenure. Figure 3 shows life-cycle income, wealth, and homeownership by income group.

Detailed interpretation: Figure 2 is central for understanding spatial misallocation from housing policy. If RS moves lower-productivity households into Manhattan, the model can deliver insurance gains but also output losses from weaker agglomeration sorting. Figure 3 shows why policy effects differ by age and wealth: older and wealthier households are homeowners with capital gains or losses, while younger and poorer households care more about rent stability.

Economic message: The model links spatial sorting to life-cycle wealth. Housing policy is simultaneously redistribution, insurance, and spatial reallocation.

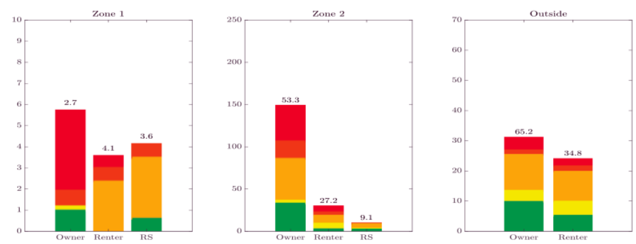


FIGURE 2

Geographic distribution of households by productivity

Notes: The colours indicate productivity levels. For working-age households: red indicates a top 12.5% productivity household, brown a household in the next 12.5% of the productivity distribution, okra: a household in the middle 50% of productivity, and yellow a household in the bottom 25%. Retired households of all productivity levels are indicated by green. The vertical axes measures the total square footage devoted to the various types of *housing* in each zone. Numbers reported atop each of the six vertical bars are the percentage of *households*; they sum to 100% across the six housing status categories in Zone 1 and Zone 2, and they sum to 100% across the two housing status categories in the outside MSA.

(a) Figure 2: Geographic distribution of households by productivity. Stacked bars show how productivity types and retirees occupy owner, renter, and RS housing in each location.

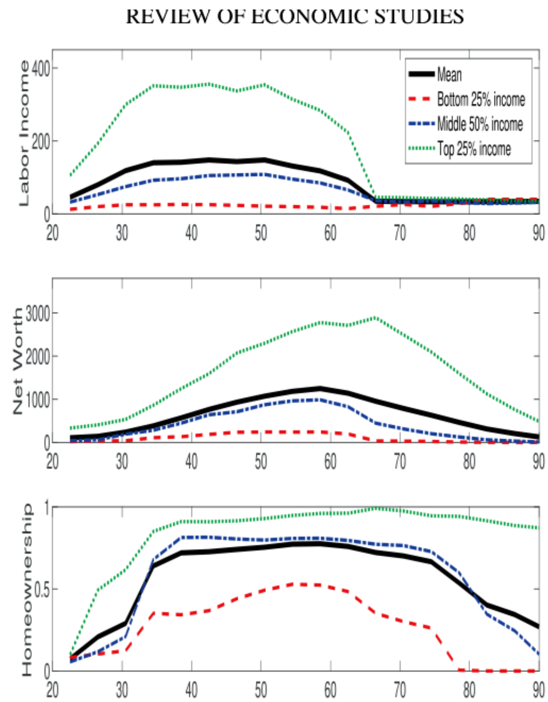


FIGURE 3

Income, wealth, and home ownership over the life cycle

e model-implied income distribution (top row), wealth distribution (middle row), and hon
/ York metropolitan area. The different lines in each panel refer to the different income

(b) Figure 3: Income, wealth, and home ownership over the life cycle. The panels show how the calibrated model generates life-cycle inequality and tenure patterns.

6.3 Figure 4: Prevalence of RS

Read: Figure 4 plots the share of households in RS by age and income group in the model and in data.

Detailed interpretation: RS is not a perfectly targeted transfer. It reaches many low-income households but also many middle- and high-income households because allocation is by lottery and continued eligibility is not income-tested. This imperfect targeting motivates the income-qualification reform.

Economic message: Rent stabilization provides durable insurance, but the baseline system is not sharply targeted to the neediest households.

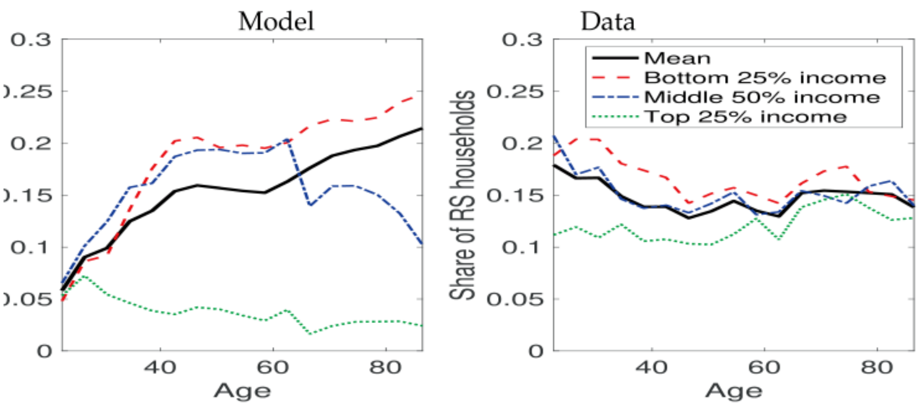


FIGURE 4

Prevalence of RS

he figure plots the share of households in rent stabilized rental housing units out of all housing units. Age is on the horizontal axis. age, we split households into the bottom-25% of income, the middle 50%, and the top-25%. The results for the model are plotted ft. The results from the data are plotted on the right. Since RS status by age and income is only available from the New York City and Vacancy Survey, the data only pertains to the five counties of New York City rather than to the full MSA. For the purposes of h only, we include rent-controlled units in the numerator of the RS share. The shares are rescaled to deliver the overall RS share tire MSA.

Figure 4: Prevalence of RS. The model and data panels show how RS coverage varies over the life cycle and across income groups.

6.4 Table 2: Policy experiments: main moments

Read: Column (1) expands RS by 50% and raises welfare by 0.91%. Column (2) adds income qualification and yields 0.66%. Column (4) upzones Zone 1 and yields 0.11%. Column (5) expands vouchers and yields approximately zero. Column (6) implements a progressive cash transfer and yields 3.90%.

Detailed interpretation: Table 2 is the main policy comparison. The key lesson is not simply rent control good or bad. The trade-off is insurance versus distortions. RS expansion works because it provides stable insurance to many low-income households; vouchers fail because tax financing and migration responses magnify financing costs; upzoning is broadly positive but less redistributive; targeted cash transfers are powerful because they deliver insurance without forcing housing misallocation.

Economic message: Policy design and financing are first order. Housing policies are evaluated by how well they target insurance while minimizing supply, labour, and migration distortions.

TABLE 2
Policy experiments: main moments

		Benchm.	(1) RS share 1.50× (%)	(2) Inc Qual Stay (%)	(3) All RS in Z2 (%)	(4) Zoning (%)	(5) Vouchers (%)	(6) Cash transfer P (%)
1	Avg(rent/inc.) renters in Z1 (%)	39.5	−0.58	14.00	−4.74	0.28	−0.36	−1.39
2	Avg(rent/inc.) renters in Z2 (%)	32.8	−2.58	11.24	−0.87	0.63	0.79	4.10
3	Fraction of HHs in RS (%)	14.20	64.47	29.23	–	0.14	4.44	5.11
4	Frac. in RS of those in inc. Q1 (%)	16.48	58.72	105.06	22.43	0.16	6.18	5.98
5	Frac. rent-burdened (%)	54.7	3.97	22.28	−3.96	−0.18	3.88	4.12
6	Avg. size of RS unit in Z1 (sf)	869	−3.82	−48.65	–	0.19	−0.17	−0.29
7	Avg. size of RS unit in Z2 (sf)	862	−1.27	−38.04	−0.13	0.02	−1.76	−1.86
8	Avg. size of a Z1 mkt unit (sf)	1,156	−6.54	−2.52	−3.30	0.40	−1.13	−0.77
9	Avg. size of a Z2 mkt unit (sf)	1,824	5.50	−2.61	−2.05	−1.04	−4.61	−4.99
10	Frac. of pop. living in Z1 (%)	10.5	14.09	10.61	−8.43	6.95	−1.95	−3.01
11	Frac. of retirees living in Z1 (%)	17.5	22.88	114.80	−34.48	4.14	3.13	0.65
12	Housing stock in Z1	–	0.36	0.37	0.62	8.80	−0.50	−0.20
13	Housing stock in Z2	–	−0.25	1.17	−0.65	−0.62	−2.69	−1.86
14	Rent/sf Z1 ($)	3.55	12.03	6.94	−1.55	−0.29	1.79	2.38
15	Rent/sf Z2 ($)	1.28	3.21	7.48	1.45	0.24	1.80	2.44
16	Price/sf Z1 ($)	884	11.28	6.82	−1.38	−0.27	1.72	2.41
17	Price/sf Z2 ($)	249	3.17	7.65	1.48	0.23	1.69	2.45
18	Homeownership rate in Z1 (%)	34.8	−116.32	−6.26	54.26	8.30	4.26	5.09
19	Homeownership rate in Z2 (%)	61.1	−10.75	−2.62	−6.30	−0.64	−6.13	−8.03
20	Avg. inc. Z1 working-age HHs ($)	167,840	−42.47	−17.02	29.20	2.70	4.60	5.83
21	Avg. inc. Z2 working-age HHs ($)	99,755	6.69	−0.64	−6.39	−1.31	−2.09	−2.26
22	Frac. of top-prod HHs in Z1 (%)	21.6	−82.31	4.73	32.35	10.10	5.85	6.07
23	Total hours worked	–	1.58	6.69	1.69	0.76	0.56	1.56
24	Total hours worked in effic. units	–	0.02	−0.99	−0.06	0.28	−3.48	−2.69
25	Total output	–	0.03	−0.63	−0.08	0.16	−2.24	−1.73
26	Total commuting time	–	0.49	9.18	2.36	0.05	1.10	2.28
26	Developer profits	–	−0.55	3.10	−0.24	0.21	−3.37	−2.78
27	Access to RS insurance (%)	4.1	68.83	94.73	13.55	1.24	2.42	8.28
28	Stability of RS insurance (%)	79.9	0.21	2.03	0.90	−0.10	0.62	−0.06
29	Std. MU growth, non-durables	0.60	−1.21	2.80	3.42	0.78	2.73	0.41
30	Std. MU growth, housing	0.60	−4.14	3.51	1.42	1.13	1.45	−1.09
31	**Aggr. welfare change (NY pop)**	–	**0.91**	**0.66**	**0.25**	**0.11**	**−0.00**	**3.90**
32	NY population	7,124.9	1.89	9.90	2.66	1.18	2.12	3.06
33	Aggr. welfare change (no migr.)	–	1.36	0.32	0.37	0.40	0.53	4.66

Notes: Column "Benchmark" reports values of the moments for the baseline model

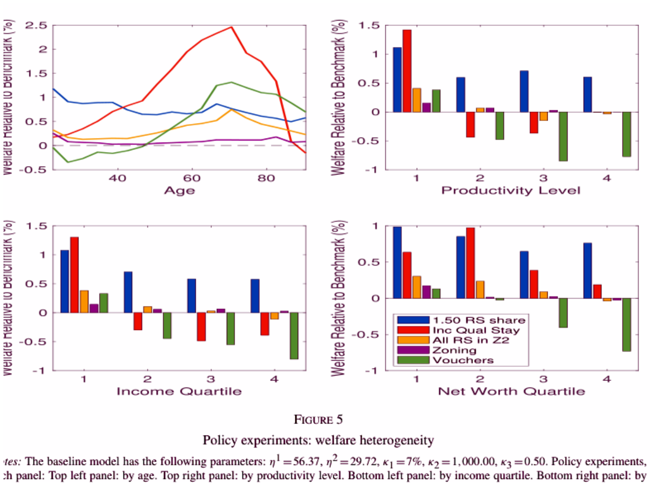
Table 2: Policy experiments: main moments. The table compares benchmark outcomes to six counterfactual policy experiments across affordability, housing supply, prices, labour, migration, insurance, and welfare.

6.5 Figure 5: Policy experiments: welfare heterogeneity; Figure 6: Policy experiments: migration

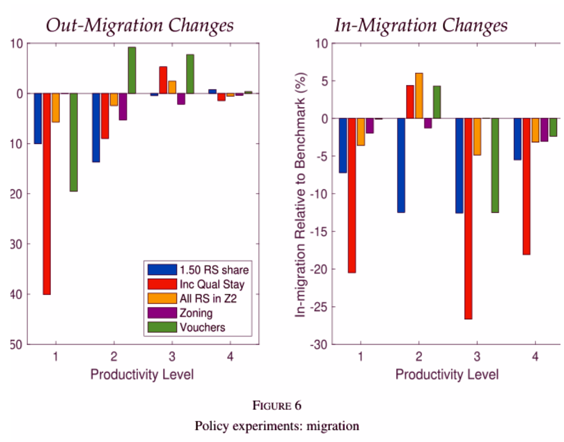
Read: Figure 5 decomposes welfare effects by age, productivity, income, and net worth. Figure 6 shows out-migration and in-migration changes by productivity.

Detailed interpretation: These figures translate the aggregate welfare numbers in Table 2 into incidence and mobility. Figure 5 shows who wins and loses; Figure 6 shows how city composition changes. These patterns are crucial because average welfare is driven by high-marginal-utility households, while migration affects output and the tax base.

Economic message: The best housing policy is not the one with the largest average rent reduction; it is the one that improves insurance for vulnerable households without inducing large migration or productivity distortions.



(a) Figure 5: Policy experiments: welfare heterogeneity. Bars and lines show distributional effects by age, productivity, income, and wealth.



(b) Figure 6: Policy experiments: migration. The panels show how each policy changes out-migration and in-migration by productivity type.

7 Short summary

- The paper studies housing affordability policy in a dynamic stochastic spatial equilibrium model.
- The model is calibrated to New York and features Manhattan, the rest of the MSA, and an outside metro.
- Households face income and mortality risk and choose labour supply, savings, tenure, housing, location, and migration.
- Rent stabilization provides insurance but distorts supply, turnover, housing quantities, and spatial sorting.
- Expanding RS raises welfare by 0.91%; income-qualifying RS raises welfare by 0.66%; upzoning raises welfare by 0.11%; vouchers generate zero average welfare; progressive cash transfers generate 3.90%.
- The central lesson is that housing affordability policy must be evaluated in general equilibrium with insurance, migration, and distributional effects.

8 Conclusion

8.1 Core conclusion

Affordable housing policy can generate large welfare gains when it provides insurance to low-income households in an incomplete-markets city. The result is strongest for policies that expand or better target rent stabilization. Housing policy is costly, but in the calibrated New York economy the insurance value can dominate distortions.

8.2 Economic intuition

Housing is both a consumption good and a risky long-term commitment. Low-income renters face income shocks, uncertain future rents, and limited access to financial insurance. A rent-stabilized unit stabilizes a large part of the household budget and therefore stabilizes marginal utility. The cost is that rent discounts distort housing consumption, reduce turnover, weaken developer incentives, and move households across space.

8.3 Policy implications

- Rent stabilization can be welfare-improving when it expands insurance access for vulnerable households.
- Income qualification improves targeting and reduces misallocation.
- Upzoning is useful and broadly beneficial, but its average welfare effect is modest when it does not directly insure high-marginal-utility households.
- Voucher programs need careful financing and design; otherwise migration and tax-base responses can undo their direct benefits.
- Highly progressive cash transfers can dominate housing-specific policies in the model, but may be harder for local governments to implement.

8.4 Limits and interpretation

The model is calibrated rather than estimated by a quasi-experimental design. Results depend on assumptions about risk aversion, moving costs, housing supply elasticities, zoning limits, RS allocation, and tax financing. Still, the model is valuable because it forces all major margins—prices, construction, migration, labour supply, and distribution—to clear jointly.

8.5 Takeaway

The paper moves the debate from partial-equilibrium claims about rent control toward a full welfare accounting. Affordable housing policies should be judged by whether their insurance value to high-marginal-utility households exceeds their supply, labour, spatial, and fiscal distortions.